

Context Statement

This memo aims to retrospectively examine natural disaster relief management with a particular focus on equity. Specifically, by analyzing the distribution of grants awarded following Hurricane Katrina in 2005, our aim is to help reform programs to better serve those in need. This memo aims to inform disaster aid administrators, who ultimately make decisions on the recipients and amounts associated with grant funding. In particular, with the memo, we address the director of the Louisiana Office for Community Developments, whose department is responsible for leading rebuilding efforts following natural disasters in the state of Louisiana, a hurricane-prone state. Our research is focused around the following two questions:

- What proportion of total property damage is recouped in disaster relief funding, and how does this vary by income status?
- What is the average amount of excess benefits awarded to citizens, and how does this total vary between insured and uninsured individuals?

By answering the questions above, we aim to spotlight and correct the inequity in relief aid, where lower-income populations are receiving inequitable treatment as they attempt to rebuild. This helps serve as a case study to rethink and retool disaster relief distribution such that vulnerable populations are prioritized in future recovery efforts.

Executive Summary / TL;DR

Following hurricanes Katrina and Rita in 2005, Louisiana launched the Road Home Program to help homeowners rebuild or repair storm-damaged property. However, with controversies in equitable allocation, this case study offers valuable insights into future program design. Learning from the mistakes of previous allocation mishaps, our study uncovers income disparities in aid allocation which should be resolved in future programs in order to provide fair, unbiased, and equitable outcomes for all residents.

By analyzing the state's own grant data, we demonstrate that the distribution of disaster relief funding was regressive, aiding homeowners with alternative means of aid, such as insurance payouts, rather than prioritizing homeowners without alternative means of aid. This leads to excess funding awarded to those who do not need it, and takes away timely funding from those who may need it most. We also demonstrate that the program's reliance on pre-storm property values, rather than actual damage assessments, systematically disadvantaged low-income homeowners whose properties were worth less before the storms.

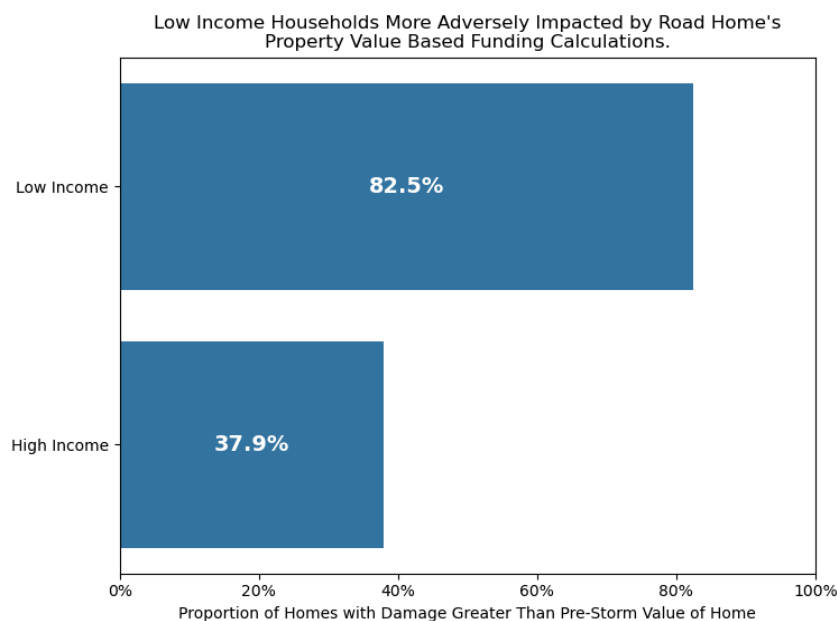
To prevent perpetuating inequalities during emergency situations, we advise disaster relief administrators to use damage estimates rather than property values when calculating aid distributions. Furthermore, we would advise administrators to better coordinate with insurance funding to eliminate wasteful, excess benefits and the redirection of such funding to address disproportionate gaps in coverage of damages for lower-income residents.

Louisiana Underfunded Poorer Populations in Recovery Efforts

In Louisiana, one of the most hurricane-prone states in the U.S, the equitable distribution of relief funds is not just a matter of fairness, but one of long-term community resilience. When aid is distributed inequitably, recovery slows for vulnerable populations and additional burden is placed on state and local agencies to fill unmet needs. Therefore, we seek to understand if Louisiana's Road Home Program equitably provided compensation to offset the amount of damage incurred by residents.

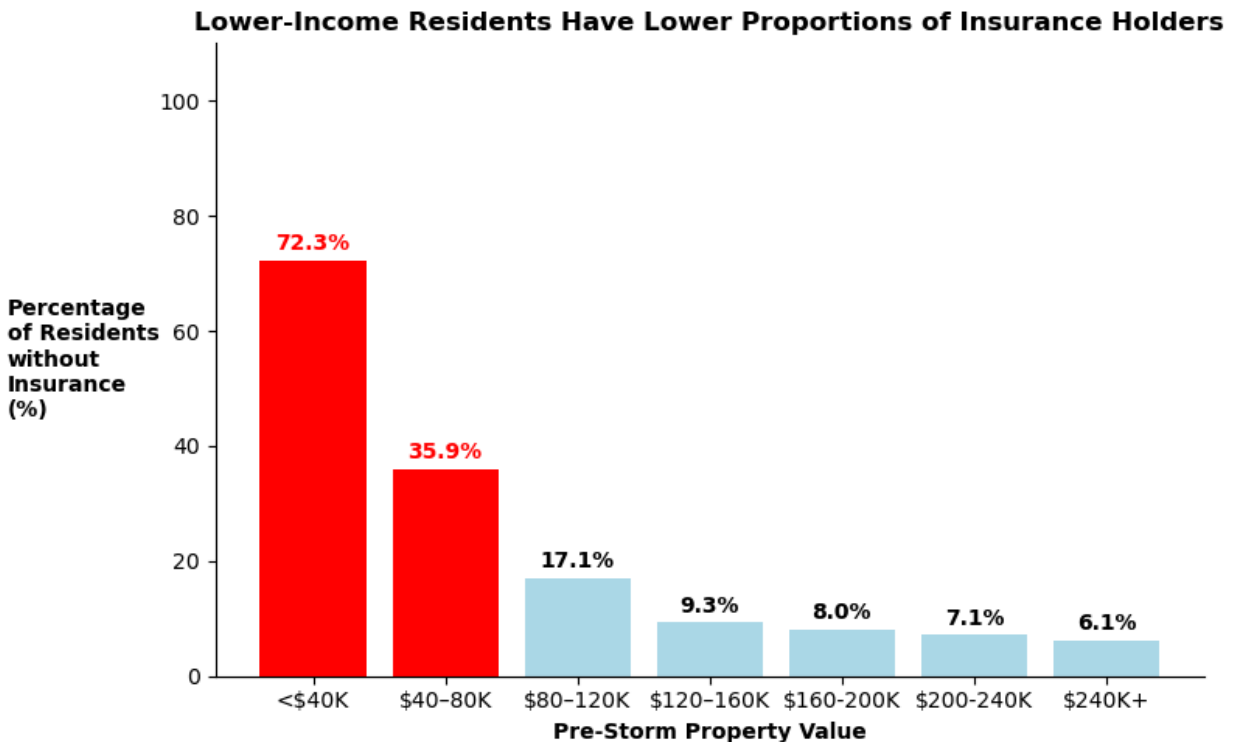
By examining how the Louisiana Road Home Program, a program set up to help homeowners rebuild or repair storm-damaged property, allocated grants after Hurricanes Katrina and Rita, our research found that lower property value homeowners had a lower proportion of their hurricane damage covered by the Road Home grants when compared to homeowners of property with higher property value. The program's grant formula was based on the lower of a property's pre-storm value or its damage assessment, meaning that poorer residents with low property value received less money than necessary to cover damages, even when their homes were completely destroyed.

As the chart below illustrates, 82.5% of low-income households suffered damages that exceeded the value of their homes, compared to 37.9% of higher income households, leaving low-income households with significantly less funding to rebuild. Since funding calculations used pre-storm home values, higher-value homeowners often had damages that were less than their comparatively higher pre-storm property values, allowing their grants to fully cover repair costs while leaving poorer families unable to fully cover damages. Future disaster relief administrations should pivot away from this grant model based on pre-storm value and reference damages instead in order to equitably help residents rebuild and recover.



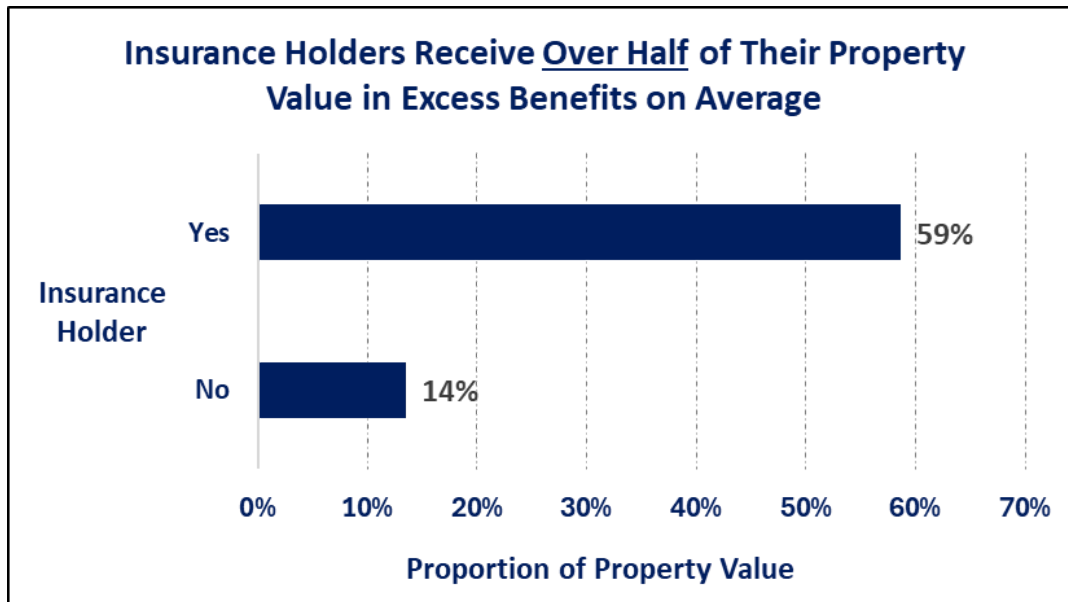
Residents with Insurance Policies Receive Excess Benefits

The insufficiency of funding to cover damages for lower-income residents only captures half the inequality story. To obtain the full picture and rectify the inequality gap in program payouts, it is necessary to analyze payouts awarded to wealthier residents. Our analysis pivots to the other half, and finds that wealthier residents received excess funding from the Road Home grant program due to duplicative benefits from the Road Home program and private insurance. The below graph illustrates that fewer residents with higher property values lacked flood and storm insurance when compared to their counterparts with lower property values.



Given that the proportion of non-insured individuals is significantly higher for populations with lower property value compared to those with higher property value, such policies stood to generate an outsized gap in disaster relief. With fewer lower-income residents enrolled in storm and flood insurance, our study also found, on average, those with insurance received almost six times the duplicative benefits, or excessive grant money, compared to residents without insurance.

The below chart shows that government programs did not account for insurance payouts, leading to insurance holders enjoying over 50% excess benefits compared to their pre-storm property values. For future policy design, this illustrates that government programs should coordinate with insurance agencies on total payout to ensure sufficient need exists for funds. Additionally, when combined with the prior finding that 82.5% of lower-income residents were uncompensated for total damage based on a grant formula tied to property value, such funds should be prioritized to assist residents with unmet damages rather than excessively compensating those whose insurance policies cover such damage.



Takeaways

Our findings highlight that the Road Home Program's reliance on pre-storm property values systematically disadvantaged low-income homeowners. Because grant amounts were capped by the lower of property value or damage, homeowners with less valuable properties received less assistance—even when their homes experienced higher amounts of damage.

Therefore, we recommend that future disaster relief programs base grant calculations on verified damage assessments rather than pre-storm property values. This approach ensures that aid aligns with actual losses, not with preexisting economic disparities.

Furthermore, our analysis shows that many higher-income households received excess funding due to overlapping sources of assistance, such as insurance payouts and federal aid. This inefficiency not only misdirects limited funds but also contributes to delays for households still waiting for sufficient support.

To prevent duplicative payments and redirect excess resources toward those with unmet needs, we recommend improved coordination between state disaster agencies and private insurers. Such coordination would allow relief programs to identify overlapping aid in real time and reallocate funds where they are most needed to cover damage.

Taken together, these adjustments would help future rebuilding efforts prioritize need over asset value, leading to more equitable recovery outcomes and stronger, more resilient communities across Louisiana's Gulf Coast.